

Radical Equations, Domains, and Decimals

Answer Key

Algebra 2

Quick Answers

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|---|--|
| 1. (a) domain = $[5, \infty)$, (b) simplified form = $2\sqrt{x-5}$ | 7. 11.08 |
| 2. 38 | 8. 6 |
| 3. 2.84 s | 9. $(4, \infty)$ |
| 4. 2 | 10. 0.99 m |
| 5. $(-\infty, \frac{7}{2}]$ | 11. $\frac{1}{4}$ |
| 6. 4 | 12. (a) domain of the radical = $[-\frac{7}{2}, \infty)$, (b) the solution = 1, (c) left side at $x = -3 = 1$, — |
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What is verified

15 of 16 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.A.2 – problems 2, 4, 6, 7, 8, 11, 12

HSN-RN.A.2 – problems 1, 3, 5, 9, 10, 12

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

4. If they got 2, -1: kept the candidate $x = -1$ that the squaring step introduced
8. If they got 6, -1: reported both roots of the squared equation, $x = 6$ and $x = -1$
12. If they got 1, -3: kept both roots of the squared equation, $x = 1$ and $x = -3$
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1. $f(x) = \sqrt{x-5}$. (a) Domain. (b) $\sqrt{4(x-5)}$ as $k\sqrt{x-5}$.

Solution (a): The radicand must be non-negative: $x - 5 \geq 0$.

$$\boxed{x \geq 5}$$

Solution (b): 4 is a perfect square and it multiplies the radicand, so the root does come apart over that product: $\sqrt{4(x-5)} = \sqrt{4} \sqrt{x-5}$.

$$\boxed{2\sqrt{x-5}}$$

2. Solve $\sqrt{x-2} = 6$.

Solution: Square both sides: $x - 2 = 36$. Check: $\sqrt{38 - 2} = \sqrt{36} = 6 \checkmark$

$$x = 38$$

3. Pendulum period $T = 2\pi\sqrt{L/9.8}$ seconds, $L = 2$ metres.

Solution: Substitute and evaluate inside out:

$$T = 2\pi\sqrt{\frac{2}{9.8}} = 2\pi\sqrt{0.204081\dots} = 2\pi(0.451754\dots) = 2.83845\dots$$

Rounded to two decimal places,

$$2.84 \text{ s}$$

4. Solve $\sqrt{x+2} = x$.

Solution: Square: $x+2 = x^2$, so $x^2 - x - 2 = 0$ and $(x-2)(x+1) = 0$. Candidates $x = 2$ and $x = -1$. Test in the original: $\sqrt{4} = 2 \checkmark$; $\sqrt{1} = 1 \neq -1 \times$ (a square root is never negative).

$$x = 2$$

5. Domain of $g(x) = \sqrt{7-2x}$.

Solution: Require $7-2x \geq 0$, so $7 \geq 2x$. Dividing by the positive 2 keeps the direction of the inequality.

$$x \leq \frac{7}{2}$$

6. Solve $\sqrt{5x-4} = \sqrt{2x+8}$.

Solution: Both sides are non-negative wherever they are defined, so squaring introduces nothing: $5x-4 = 2x+8$, so $3x = 12$. Check: $\sqrt{16} = \sqrt{16} \checkmark$

$$x = 4$$

7. Solve $\sqrt{2x-1} = 4.6$ to two decimal places.

Solution: Square both sides, then isolate x :

$$2x - 1 = 4.6^2 = 21.16, \quad 2x = 22.16, \quad x = 11.08.$$

Check: $\sqrt{2(11.08) - 1} = \sqrt{21.16} = 4.6 \checkmark$

$$11.08$$

8. Solve $\sqrt{5x+6} = x$.

Solution: Square: $5x+6 = x^2$, so $x^2 - 5x - 6 = 0$ and $(x-6)(x+1) = 0$. Candidates $x = 6$ and $x = -1$. Test: $\sqrt{36} = 6 \checkmark$; $\sqrt{1} = 1 \neq -1 \times$.

$$x = 6$$

9. Domain of $p(x) = \frac{1}{\sqrt{x-4}}$.

Solution: The radicand must be non-negative, and the radical sits in a denominator so it must not be zero either. Both conditions together give $x-4 > 0$ — a strict inequality, unlike problem 1.

$$x > 4$$

10. $L = 9.8 \left(\frac{T}{2\pi} \right)^2$ metres, $T = 2$ seconds.

Solution: Evaluate the bracket first, then square:

$$L = 9.8 \left(\frac{2}{2\pi} \right)^2 = 9.8(0.318309\dots)^2 = 9.8(0.101321\dots) = 0.99295\dots$$

Rounded to two decimal places,

0.99 m

11. Solve $\sqrt{x+6} = \sqrt{x} + 2$.

Solution: Square both sides, keeping the cross term:

$$\begin{aligned}x + 6 &= x + 4\sqrt{x} + 4 \\2 &= 4\sqrt{x} \\\sqrt{x} &= \frac{1}{2} \\x &= \frac{1}{4}.\end{aligned}$$

Check: $\sqrt{6.25} = 2.5$ and $\sqrt{0.25} + 2 = 0.5 + 2 = 2.5$ ✓

$x = \frac{1}{4}$

12. Synthesis: $\sqrt{2x+7} = x+2$. (a) Domain. (b) Solve. (c) Left side at $x = -3$. (d) Why $x = -3$ fails.

Solution (a): $2x + 7 \geq 0$ gives $2x \geq -7$.

$x \geq -\frac{7}{2}$

Solution (b): Square: $2x + 7 = x^2 + 4x + 4$, so $x^2 + 2x - 3 = 0$ and $(x+3)(x-1) = 0$. Candidates $x = 1$ and $x = -3$. Test $x = 1$: $\sqrt{9} = 3 = 1 + 2$ ✓. Test $x = -3$: $\sqrt{1} = 1$ but $-3 + 2 = -1$ ✗.

$x = 1$

Solution (c): $\sqrt{2(-3) + 7} = \sqrt{1}$.

1

Solution (d): *Instructor-judged.* There are two separate tests and $x = -3$ passes only the first. It is inside the domain, since $-3 \geq -\frac{7}{2}$, so the left side is perfectly well defined and equals 1. It fails the equation because the right side $x + 2$ is -1 there, and a square root can never equal a negative number. Squaring hid that sign difference, which is why the candidate appeared at all. Full credit needs the distinction between being in the domain and satisfying the equation.